

BENEDICT E. UMEH

Senior Full Stack Engineer | AI Conversational Systems Architect | Customer Experience Technology

Leeds, UK | +44 7493 363351 | ebuka.umeh40@outlook.com
LinkedIn: linkedin.com/in/ebukaumeh | Portfolio: bhenmedia.com

PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 7+ years building production-ready, scalable applications across logistics, e-commerce, hospitality, and fintech. I specialise in AI-powered conversational systems that transform customer experience—designing and deploying multi-agent chatbots using OpenAI GPT-4o and Google Gemini with function calling, creating interactions that are warm, empathetic, naturally human, and commercially effective.

What sets me apart: deep technical engineering expertise combined with 2+ years frontline hospitality operations. I understand intimately what makes customer interactions feel natural versus robotic, how to handle escalations with empathy, and how to maintain brand voice consistency across every touchpoint. My technical foundation spans TypeScript, Python, Java, Node.js, Vue.js, React, and AWS cloud infrastructure. MSc Data Science enables rigorous, data-driven approach to CSAT improvement and continuous iteration.

TECHNICAL EXPERTISE

AI & Conversational Systems: OpenAI GPT-4o, Google Gemini API, Prompt Engineering, Function Calling, Multi-Agent AI Workflows (4+ specialized agents), Conversation Flow Design, AI Quality Assurance, Tone & Empathy Analysis, Escalation Logic, Edge Case Handling

Languages & Frameworks: TypeScript (Expert), JavaScript, Python, Java, PHP | Node.js, Express.js, Vue.js, React 18, Spring Framework, WebSockets, REST APIs

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, CI/CD (GitHub Actions), Railway, Microservices Architecture

Databases: PostgreSQL, MongoDB, MySQL, Redis, Drizzle ORM, Query Optimization, Data Modeling

Analytics & CX: Power BI, GA4, A/B Testing, CSAT/NPS Measurement, Brand Voice Consistency, Customer Journey Optimization

PROFESSIONAL EXPERIENCE

Founder & Lead AI Engineer

UK Innovator Founder Visa Assistant – AI SaaS Platform | Leeds, UK | November 2024 – Present

- Architected and launched enterprise-grade AI SaaS platform with 109+ production tools across 8 categories (Compliance, Documentation, Team, Business, Financial, Growth, Innovation, Defence), achieving 100% OISC regulatory compliance.
 - Engineered sophisticated Multi-Agent AI Chatbot System with 4 specialized conversational agents (Nova-Innovation, Sterling-Financial, Atlas-Growth, Sage-Compliance)—each with distinct personality, expertise domain, and escalation logic. Designed responses to be warm, empathetic, and naturally human while maintaining operational precision.
 - Built comprehensive AI quality assurance framework: systematically review interactions for tone, empathy, brand voice consistency, and accuracy. Maintain structured library of best-in-class responses alongside documented failure cases for continuous prompt refinement.
 - Implemented data-led approach: identify where AI fails to handle queries, update prompts accordingly, measure impact against resolution metrics. Reduced user research time by 70% through intelligent conversation design.
 - Developed real-time analytics dashboard tracking user sessions, page views, tool usage patterns, and activity heatmaps—enabling data-driven decisions on AI performance optimization.
 - Built tier-based access control system (Free/Basic/Premium/Enterprise/Ultime) with Stripe payment integration, OAuth 2.0 authentication, and session management using PostgreSQL.
 - Achieved 90+ Lighthouse performance score through frontend optimization, lazy loading, and intelligent caching strategies.
- Stack: TypeScript, React 18, Node.js, Express, PostgreSQL, Drizzle ORM, Google Gemini API, Stripe, Railway, WebSockets

Founder & Lead Software Engineer

StackRise Ltd – Enterprise SaaS Platform | Manchester, UK | December 2024 – Present

- Designed STAX AI Chatbot using OpenAI GPT-4o with function calling—conversational assistant handling complex customer queries with warmth and operational precision. Engineered to never sound robotic; every response maintains natural, human-like flow.
 - Implemented 3 CRM integration tools through AI function calling (lead collection, callback scheduling, profile updates) with validation systems to flag incorrect advice or compliance risks.
 - Built RESTful API architecture with 50+ endpoints, reducing payment completion time by 90% through Stripe integration and webhook handling.
 - Optimized API response times by 60% (800ms !' 300ms) through database query optimization and intelligent caching.
- Stack: TypeScript, Python, Node.js, PostgreSQL, OpenAI GPT-4o, Stripe, AWS, Docker, Jest

Guest Service Assistant

Ibis Styles Leeds City Centre Arena | Leeds, UK | September 2023 – Present

- Manage all night shift front-desk operations for 200+ room hotel, delivering outstanding service with consistent warmth, clarity, and genuine empathy. Handle check-ins, guest inquiries, complaint resolution, and daily reporting.
- Self-initiated AI Virtual Hotel Concierge using React and WebSockets, automating 200+ guest queries—reduced call volume by 35% while maintaining high satisfaction scores.
- Developed deep practical understanding of customer communication patterns—specifically what makes responses feel natural vs robotic, how tone affects perception, and how to handle escalations effectively.

Full Stack Developer (Contract)

Qalhata Technology – AI/Autonomous Systems | Cheshire, UK | November 2024 – Present

- Lead development for enterprise AI/autonomous systems clients, implementing end-to-end analytics pipelines and executive dashboards.

- Built automated reporting infrastructure using Python and Power BI, reducing stakeholder report turnaround from 5 days to 1 day (80% improvement).
- Mentor junior developers on quality standards, code review best practices, and Agile methodology.

Web Developer

Deskstones Ltd | Leeds, UK | February 2024 – Present

- Led corporate website rebuild achieving 45% page load improvement (3.2s !' 1.8s) through performance optimization.
- Increased SEO visibility by 30% and conversion rate by 25% through rigorous A/B testing and UX optimization.

Freelance Software Engineer & Founder

bhenmedia.com | Leeds, UK & Lagos, Nigeria | 2017 – Present

- Founded web development consultancy delivering 50+ applications for e-commerce, hospitality, healthcare, and logistics clients.
- Built e-commerce platforms with Stripe/PayPal payment processing, order tracking, and inventory management.
- Successfully migrated legacy PHP applications to modern TypeScript/Node.js architectures.

KEY ACHIEVEMENTS

- Built production multi-agent AI chatbot with 4 specialized assistants—natural, empathetic responses with proactive customer engagement
- Engineered 109+ AI-powered tools achieving 100% OISC compliance across 8 business categories
- Reduced hotel call volume by 35% through AI-powered virtual concierge with quality conversation design
- Optimized API response times by 60% through data-driven performance analysis and intelligent caching
- Delivered 50+ applications across e-commerce, logistics, healthcare, and fintech with consistent quality
- Deep frontline customer experience—understanding what makes interactions feel genuinely human and on-brand

EDUCATION

MSc Data Science | Leeds Beckett University, UK | 2023

BSc Information Technology & Business Information Systems | Middlesex University, London | 2017

CERTIFICATIONS

IBM Data Analyst Professional Certificate | Google Data Analytics Professional Certificate | Google Analytics IQ | AWS Cloud Practitioner (In Progress)